

Implementation of Centralized Log Analytics for Multi-Vector Cyber Attack Detection

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Abstract

This project presents a complete Security Information and Event Management (SIEM) solution using Docker, Node.js, Filebeat, Logstash, Elasticsearch, and Kibana. A deliberately vulnerable web application was developed to simulate common cyber attacks such as brute-force login attempts, SQL Injection, successful authentication, and unauthorized access. Application logs were collected in real time, processed through the ELK pipeline, stored centrally, and visualized through custom dashboards and detection rules. The implementation demonstrates how organizations can monitor security events, correlate attack patterns, and improve incident response using open-source technologies.

Problem Statement

Modern web applications continuously generate large volumes of logs, but without centralized monitoring these records remain underutilized and security incidents may go unnoticed. Attackers can exploit vulnerabilities such as weak authentication and insecure input handling to gain unauthorized access or extract sensitive data. Manually reviewing logs is time-consuming and ineffective when attack attempts occur at high frequency. The goal of this project is to design and implement a cost-effective SIEM environment that automatically collects application logs, detects suspicious activities such as brute-force attacks and SQL Injection, maps them to standard threat frameworks, and provides real-time dashboards and alerts for rapid investigation.

Objectives

The primary objective of this project is to build an end-to-end centralized log analytics platform that can ingest, process, analyze, and visualize security events generated by a vulnerable web application. The system is intended to demonstrate practical SIEM concepts and provide a reusable laboratory for understanding attack detection and incident monitoring.

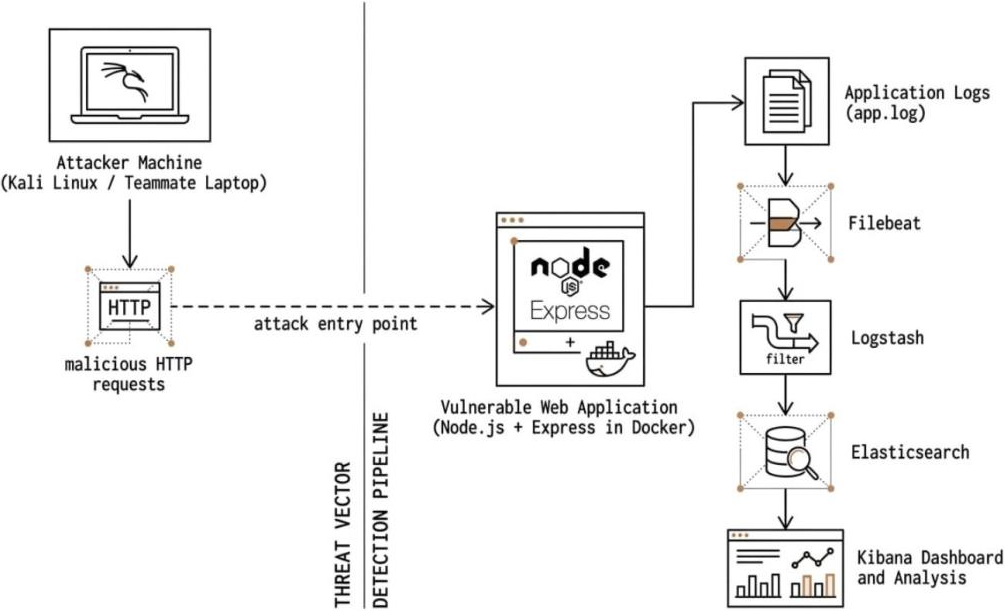
- Develop a vulnerable Node.js and Express web application for controlled attack simulation.

- Generate logs for failed logins, successful logins, SQL Injection, and unauthorized access.
- Collect raw application logs automatically using Filebeat.
- Transform and route log events through Logstash.
- Store indexed security events in Elasticsearch.
- Create Kibana Discover views and dashboards for analysis.
- Build detection rules and alerts for suspicious activity.
- Map observed attacks to MITRE ATT&CK; techniques and tactics.
- Map each attack phase to the Cyber Kill Chain.
- Demonstrate the effectiveness of open-source SIEM technologies in a lab environment.

Tools and Technologies

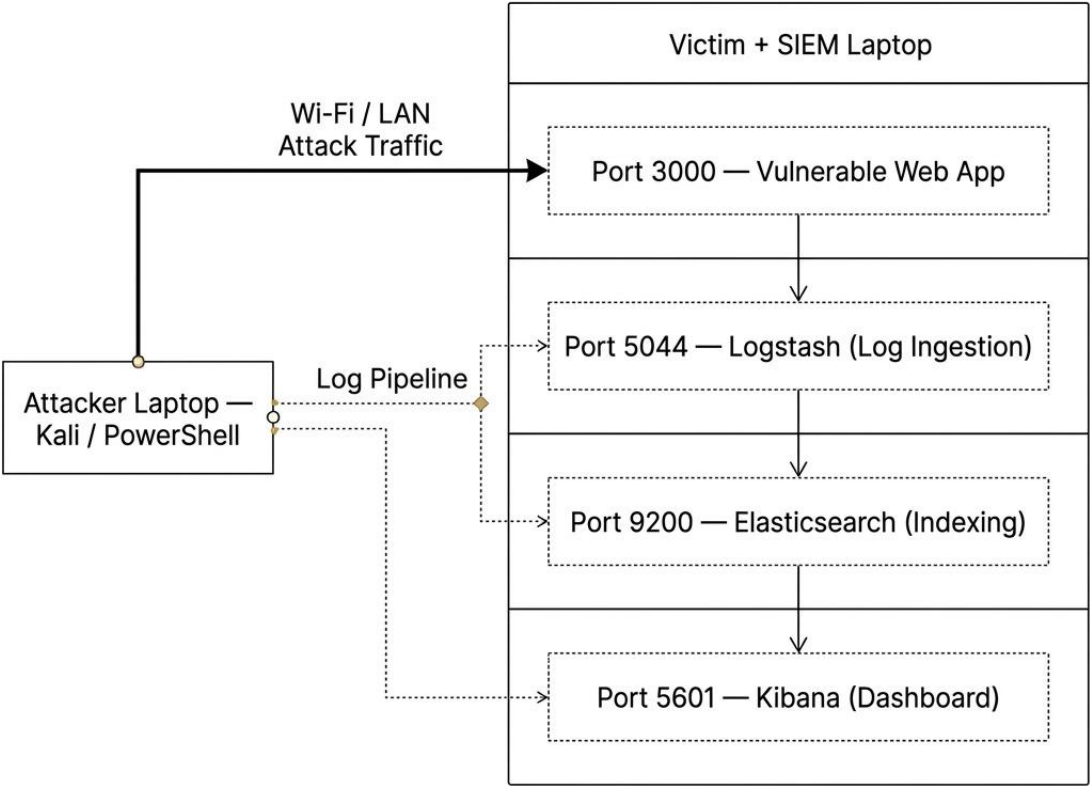
- Windows 11
- Docker Desktop
- Docker Compose
- Node.js
- Express.js
- Filebeat 7.17.0
- Logstash 7.17.0
- Elasticsearch 7.17.0
- Kibana 7.17.0
- PowerShell / Git Bash / Kali Linux

System Architecture



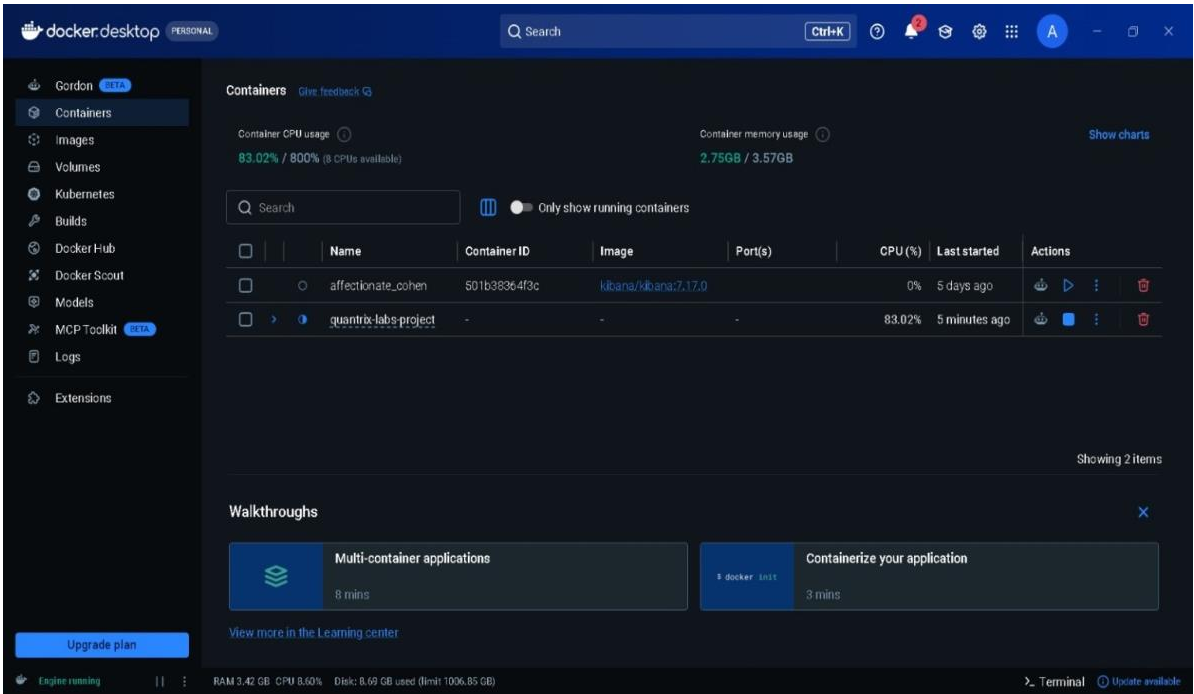
System Architecture Diagram

Network Data Flow

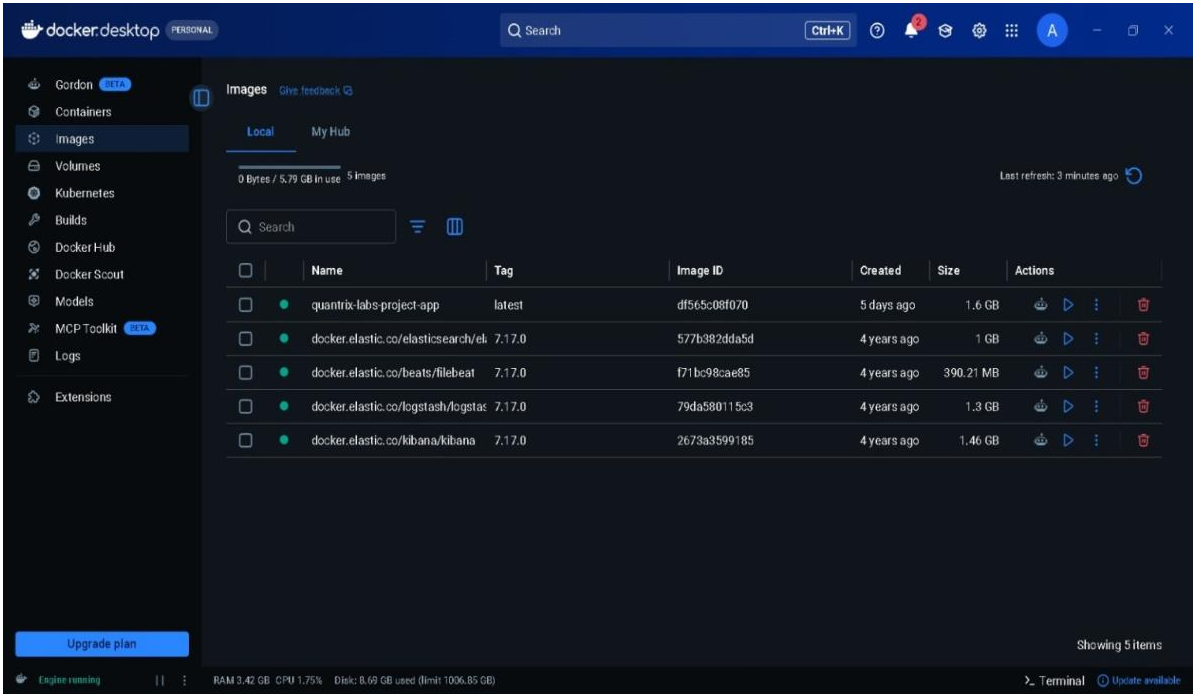


Network Data Flow Diagram

Docker Deployment



Docker Containers

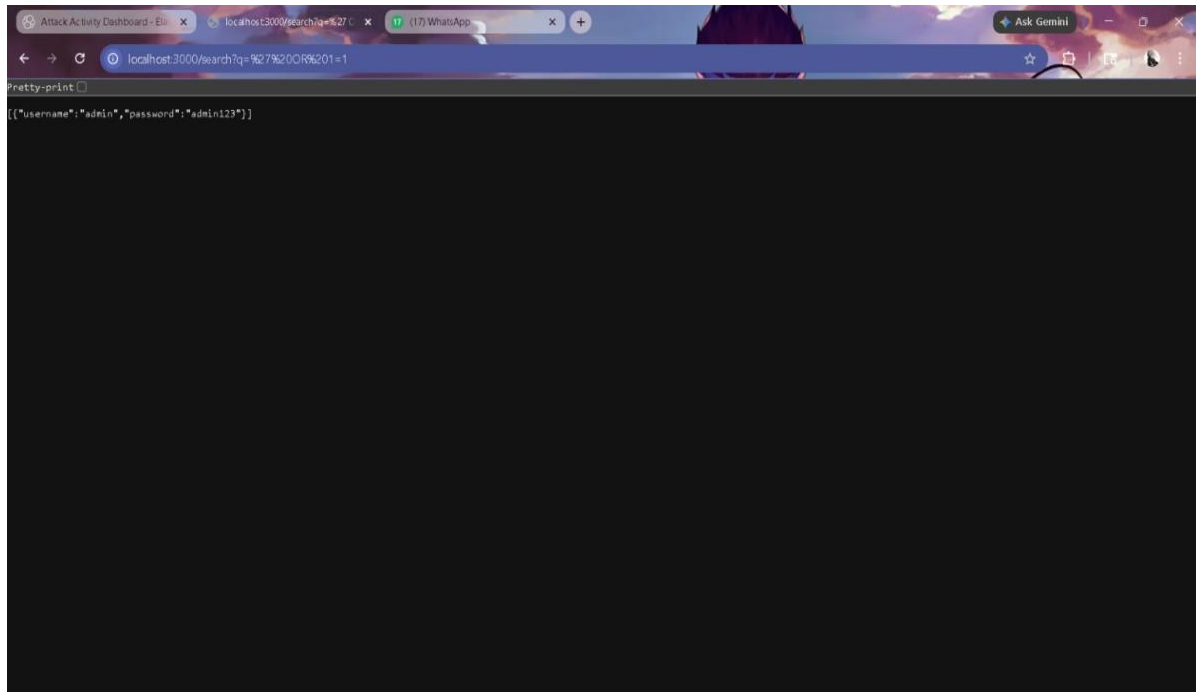


Docker Images

Attack Simulation

- Brute-force login using invalid passwords.
- Successful login using admin/admin123.
- SQL Injection using ' OR 1=1.

- Optional XSS and unauthorized admin access.



Brute Force Attack

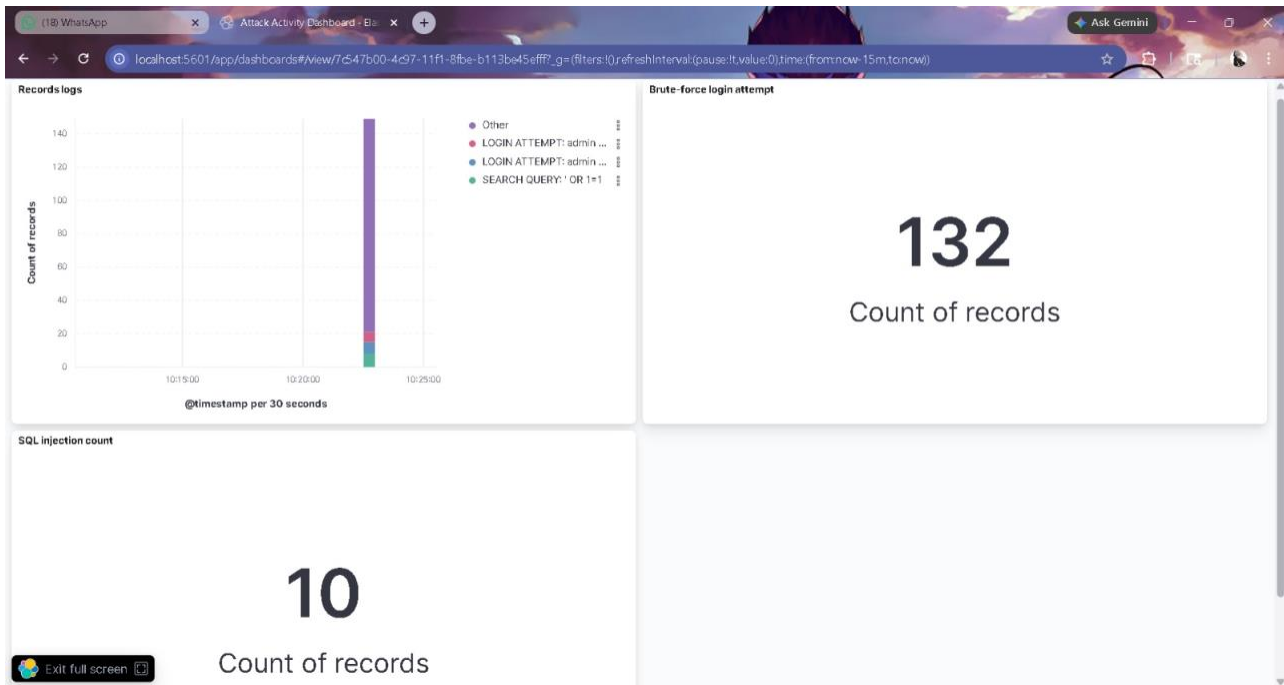
```
Windows PowerShell
C:\Users\AVINASH>curl -X POST http://localhost:3000/login -H "Content-Type: application/x-www-form-urlencoded" -d "username=admin&password=wrong11"
Invalid credentials
C:\Users\AVINASH>curl -X POST http://localhost:3000/login -H "Content-Type: application/x-www-form-urlencoded" -d "username=admin&password=wrong12"
Invalid credentials
C:\Users\AVINASH>curl -X POST http://localhost:3000/login -H "Content-Type: application/x-www-form-urlencoded" -d "username=admin&password=wrong13"
Invalid credentials
C:\Users\AVINASH>curl -X POST http://localhost:3000/login -H "Content-Type: application/x-www-form-urlencoded" -d "username=admin&password=wrong14"
Invalid credentials
C:\Users\AVINASH>curl -X POST http://localhost:3000/login -H "Content-Type: application/x-www-form-urlencoded" -d "username=admin&password=wrong15"
Invalid credentials
C:\Users\AVINASH>curl -X POST http://localhost:3000/login -H "Content-Type: application/x-www-form-urlencoded" -d "username=admin&password=wrong16"
Invalid credentials
C:\Users\AVINASH>curl -X POST http://localhost:3000/login -H "Content-Type: application/x-www-form-urlencoded" -d "username=admin&password=wrong17"
Invalid credentials
C:\Users\AVINASH>curl -X POST http://localhost:3000/login -H "Content-Type: application/x-www-form-urlencoded" -d "username=admin&password=wrong18"
Invalid credentials
C:\Users\AVINASH>curl -X POST http://localhost:3000/login -H "Content-Type: application/x-www-form-urlencoded" -d "username=admin&password=wrong19"
Invalid credentials
C:\Users\AVINASH>curl -X POST http://localhost:3000/login -H "Content-Type: application/x-www-form-urlencoded" -d "username=admin&password=wrong20"
Invalid credentials
C:\Users\AVINASH>
```

Successful Login

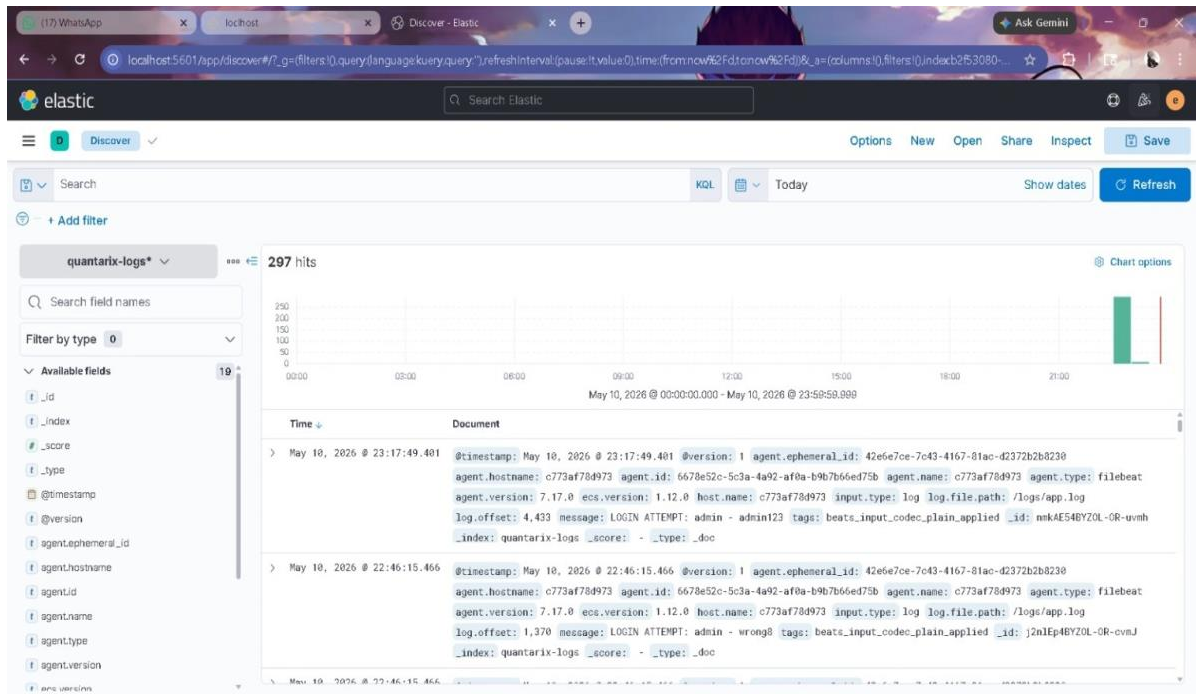
```
MINGW64:/c/Users/AVINASH
AVINASH@LAPTOP-LCMIC5K1 MINGW64 ~
$ curl -X POST http://localhost:3000/login -d "username=admin&password=admin123"
Login successful
AVINASH@LAPTOP-LCMIC5K1 MINGW64 ~
$ |
```

SQL Injection Output

Log Collection and Visualization



Kibana Discover View



Attack Activity Dashboard

Detection Rules

- Brute Force Rule: message contains "Invalid credentials".
- Successful Login Rule: message contains "Login successful".
- SQL Injection Rule: message contains "OR 1=1".

MITRE ATT&CK; Mapping

- Brute Force → T1110 (Credential Access)
- Successful Login → T1078 Valid Accounts
- SQL Injection → T1190 Exploit Public-Facing Application
- Unauthorized Admin Access → T1078 / T1190
- XSS → T1059 (Simulated Execution)

Cyber Kill Chain Mapping

- Reconnaissance
- Weaponization
- Delivery
- Exploitation

- Installation
- Command and Control (Not Applicable)
- Actions on Objectives

Results

- 296+ log records ingested.
- 132 brute-force attempts detected.
- 10 SQL injection attempts detected.
- Successful login events captured.
- Interactive dashboards and alerts created.

Conclusion

This project successfully demonstrates the implementation of a practical and fully functional SIEM environment using the ELK Stack and Docker. A vulnerable web application was used to generate realistic security events, including brute-force login attempts, SQL Injection, and successful authentications. These events were collected and processed in real time, indexed centrally, and visualized through custom Kibana dashboards. Detection rules were configured to identify malicious behavior and correlate it with MITRE ATT&CK; and Cyber Kill Chain methodologies. The project proves that open-source tools can provide powerful security monitoring capabilities at minimal cost and serves as a strong foundation for future enhancements such as automated response, advanced threat intelligence, and machine learning-based anomaly detection.

Future Enhancements

- Email and Slack notifications
- Suricata integration
- Machine learning anomaly detection
- TLS encryption and RBAC hardening